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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY

9648/01

Paper 1 Multiple Choice Questions

28 September 2016

1 hour 15 min

Additional Material: OTAS sheet

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A MCQ [40 marks]

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

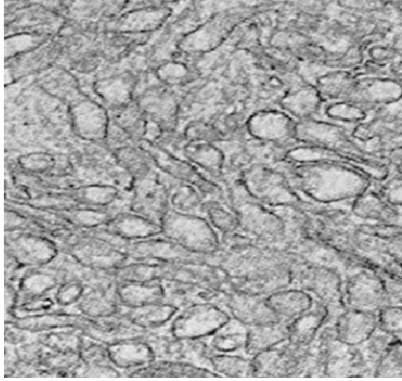
Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This document consists of **27** printed pages and **1** blank page.

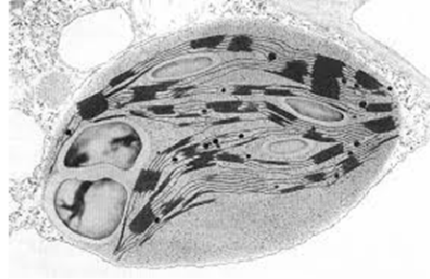
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Answer **all** questions in this section.

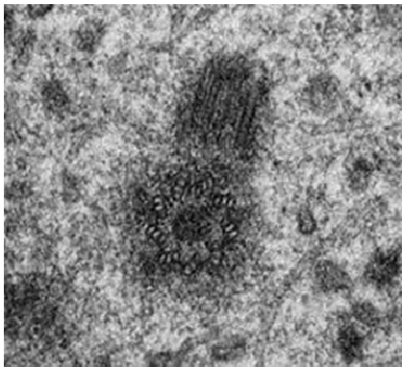
- 1 The figure below shows electron micrographs of 4 different organelles **P**, **Q**, **R** and **S**.



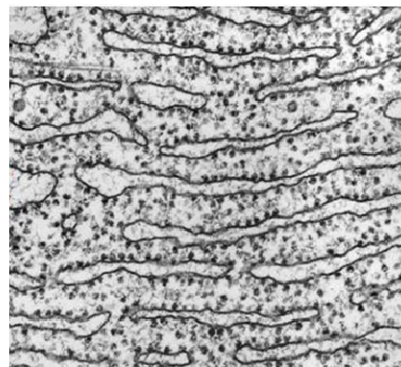
P



Q



R



S

Which of the following matches the organelle to its function?

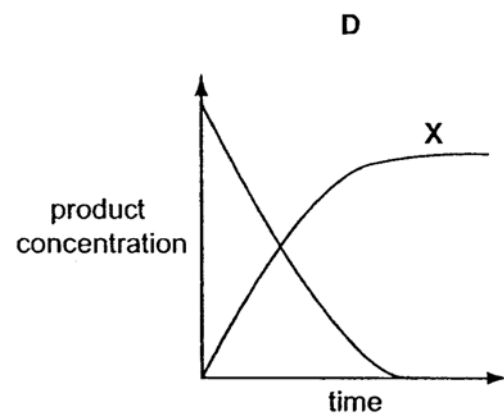
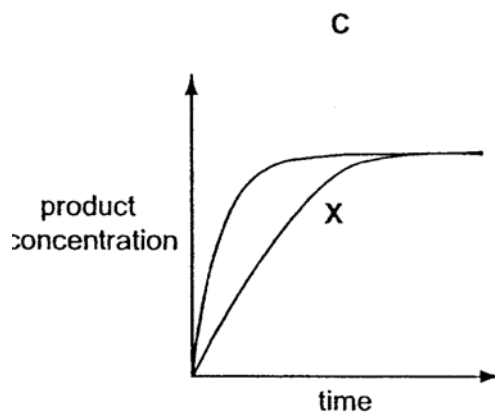
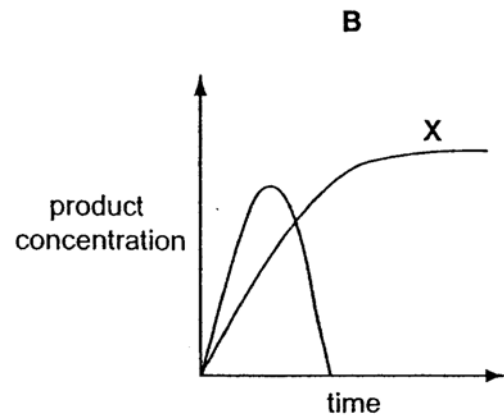
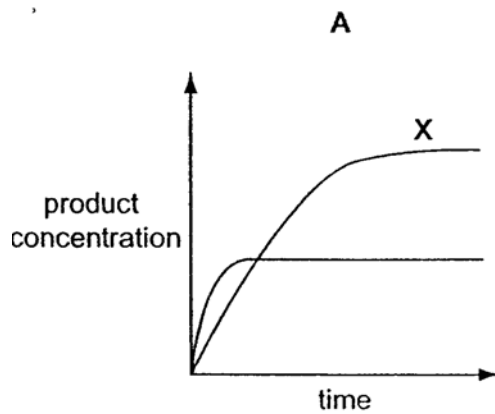
	Organelle	Function
A	P	phospholipid synthesis
B	Q	enzyme secretion
C	R	protein synthesis
D	S	glycosylation of proteins

2 Which pair shows the **CORRECT** classification?

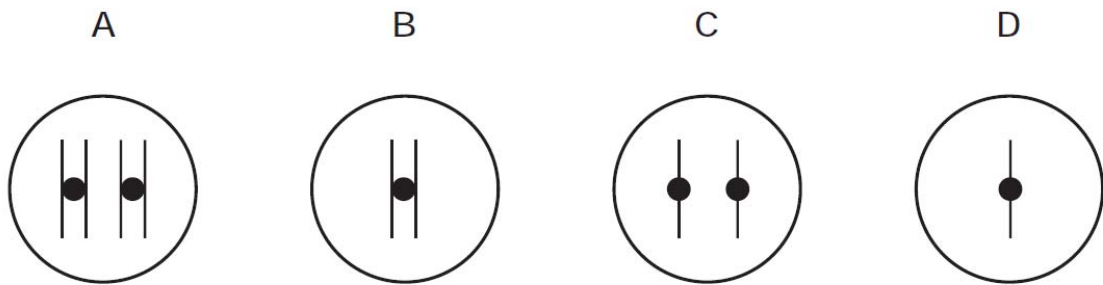
	Branched structure	Unbranched structure
A	amylose	glycogen
B	amylopectin	cellulose
C	cellulose	amylopectin
D	glycogen	amylopectin

3 Two enzyme experiments were carried out. The first, experiment X, was carried out at a constant temperature of 37°C. During the second experiment, the temperature was increased from 37°C to 80°C.

Which graph shows the results?



- 4 A cell with one pair of chromosomes ($2n = 2$) undergoes meiosis. Which nucleus is formed at the end of meiosis I?



- 5 In an attempt to synthesise DNA molecules *in vitro*, a student isolated and purified various molecules needed for DNA replication. She added some DNA to the mixture, and replication occurred. However, the DNA molecules formed were defective. Each molecule consists of a normal DNA strand paired with numerous segments of DNA, each about hundreds of nucleotides long.

What might she have left out in the mixture?

- A DNA primer
- B RNA primer
- C DNA ligase
- D DNA polymerase III

6 The mechanism of action of four drugs that inhibit DNA replication is stated below.

- **Aphidicholine** inhibits DNA polymerase III.
- **Cytarabine** is converted into a molecule that can substitute for a DNA nucleotide and also inhibits DNA repair mechanisms.
- **Epirubicin** inhibits an enzyme involved in the unwinding and separation of DNA strands.
- **Hydroxycarbamide** inhibits an enzyme involved in the production of deoxyribonucleotides.

Which row **CORRECTLY** matches the effects of these drugs on DNA replication?

	Effects of Drug on DNA Replication			
	Inhibition of chain elongation	DNA damaged during replication	DNA strands not available as templates for replication	Exposed DNA template strands unable to be copied
A	aphidicholine	hydroxycarbamide	epirubicin	cytarabine
B	cytarabine	epirubicin	aphidicholine	hydroxycarbamide
C	epirubicin	hydroxycarbamide	cytarabine	aphidicholine
D	hydroxycarbamide	cytarabine	epirubicin	aphidicholine

- 7 The diagram shows part of the normal sequence of an mRNA molecule.

CCAAGUGGUCCGCUAAGAAGGC

A mutation in the DNA resulted in a polypeptide beginning with the following sequence.

glycine - serine - proline - glycine - isoleucine - leucine

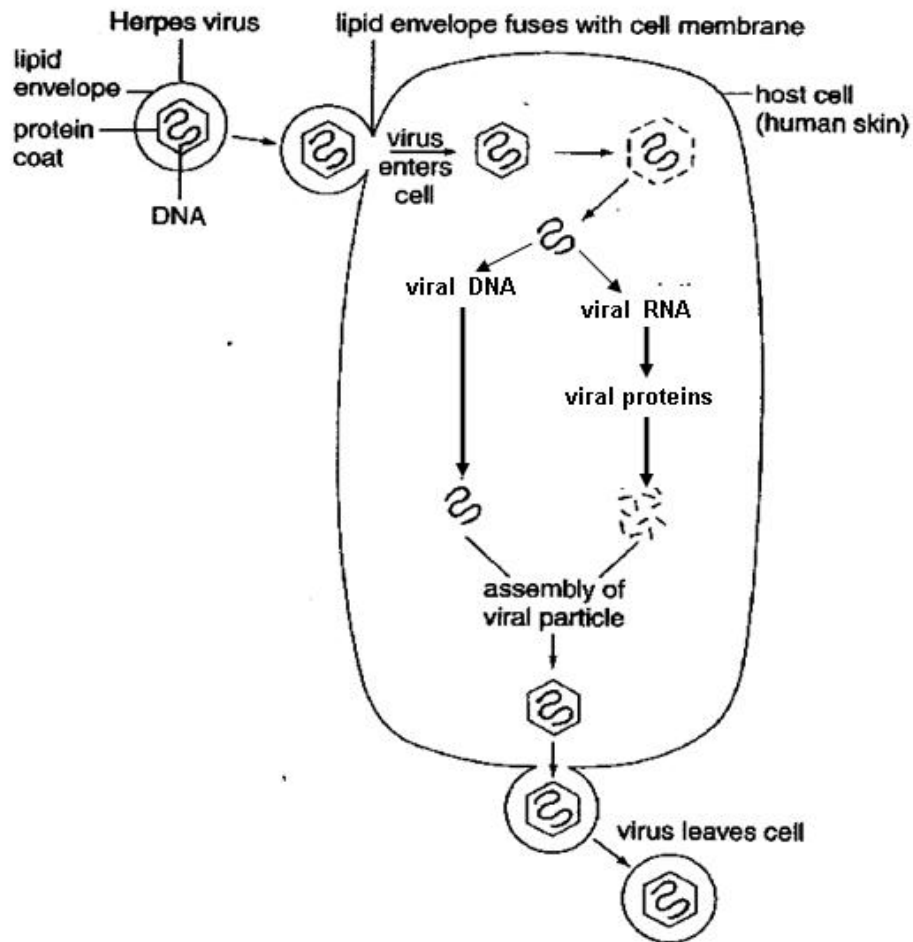
The DNA triplets for some amino acids are

Glycine	Isoleucine	Leucine	Proline	Serine
CGA	ATA	TTA	CCA	TCA
GGT	ATT	CTT	CCG	TCG
GGC		CTC		

Which mutation has occurred in the DNA molecule?

- A** The replacement of one nucleotide by a different nucleotide.
- B** A reversal in the order of nucleotides.
- C** An addition of an extra nucleotide.
- D** The loss of a nucleotide.
- 8 All of the following statements about viruses are true **EXCEPT** _____.
- A** The genome of RNA viruses are more likely to mutate than those of DNA viruses.
- B** All viruses produce RNA as an intermediate molecule during the production of new viruses.
- C** All RNA viruses produce DNA as an intermediate molecule during the production of new RNA viruses.
- D** Before entering a host cell, specific proteins of viruses bind to receptors on specific host cells.

- 9 The diagram below shows the reproductive cycle of the herpes virus which causes cold sores on the mouth.



With reference to the diagram below, which of the following statements **BEST** describes the herpes virus?

- A It is not a retrovirus as it does not contain RNA as its genetic material.
- B Its mode of replication is similar to that of influenza virus.
- C Its replication cycle includes a lysogenic phase.
- D Death of the host cell is necessary for the release of the viral progeny.

- 10 In a repressible operon under negative control, a mutation that alters the product of the operon's regulatory gene such that it is unable to bind to the co-repressor occurred.

This mutation will result in _____.

- A irreversible binding of the repressor to the operon
- B no transcription of genes of the operon
- C continuous transcription of genes of the operon
- D no difference in the transcription rate

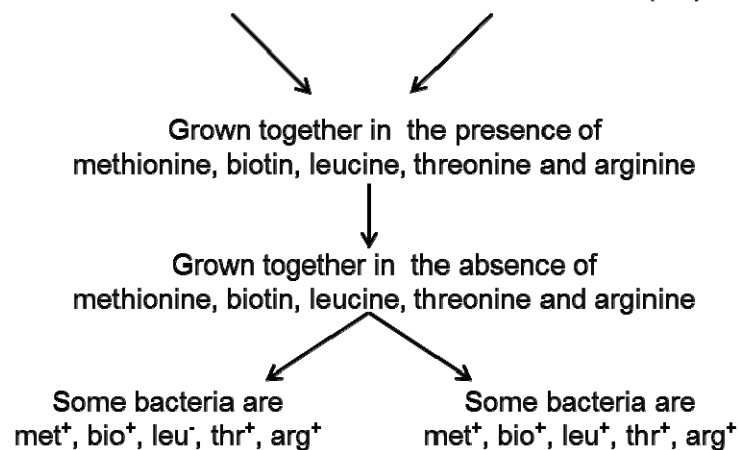
- 11 The diagram shows an investigation into bacterial genetics.

To grow mutant bacteria X:

- Need methionine (met⁻), biotin (bio⁻) and leucine (leu⁻)
- Do not need threonine (thr⁺), arginine (arg⁺)

To grow mutant bacteria Y:

- Do not need methionine (met⁺), biotin (bio⁺) and leucine (leu⁺)
- Need threonine (thr⁻), arginine (arg⁻)



Which process or processes could explain these results?

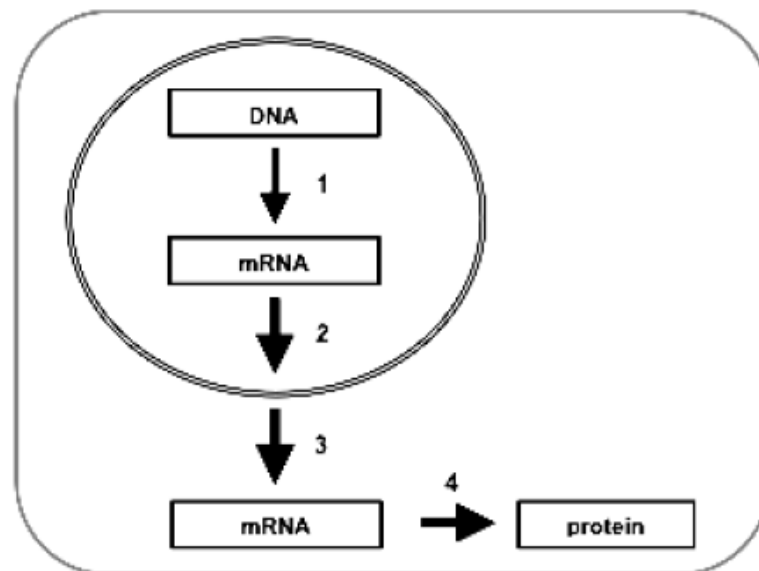
- I conjugation
- II transduction
- III transformation

- A I only
- B III only
- C I and II
- D I and III

12 Which of the following is **TRUE** of cancers?

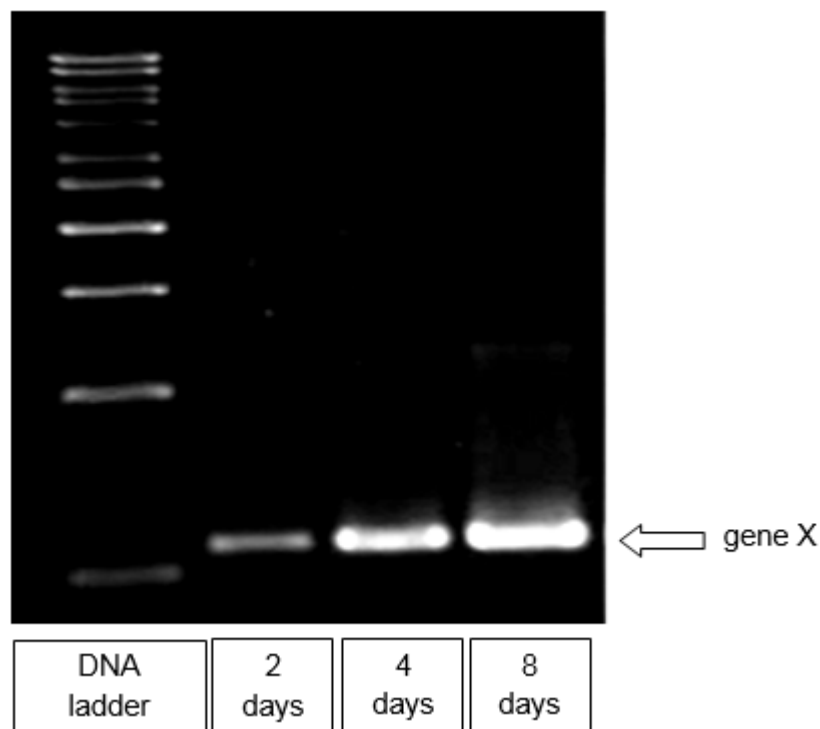
- A Anchorage dependence is lost in cancer cells.
- B Cancer cells are likely to have longer-than-usual telomeres despite having inactivated telomerases.
- C A cell that has a copy of the p53 tumour suppressor gene inactivated can be considered to be cancerous.
- D When a copy of the ras proto-oncogene is activated into an oncogene in a normal cell, cancer immediately develops.

13 The following diagram shows the expression of a particular gene to its protein product in a eukaryotic cell. Which of the following combination correctly describes steps 1 – 4?



	1	2	3	4
A	DNA is demethylated	5' capping occurs	RNase does not degrade 5' capped mRNA	Initiation factors bind to ribosome
B	DNA is demethylated	Alternative splicing occurs	Activators bind to enhancers	Ribosome binds to 5' UTR
C	DNA is methylated	Poly(A) tail is added to 3' end	poly-A tail is extended	Phosphorylation of protein
D	DNA is methylated	5' capping occurs	Removal of 5' cap	Activators bind to enhancers

- 14 Which statement best explains how related genes involved in the same metabolic pathway are expressed together in eukaryotic cells?
- A Related genes are usually located on the same chromosome so that they can be controlled by the same set of control elements.
 - B The same set of general transcription factors may be capable of recognising the same promoter site of related genes.
 - C There are specific sets of control elements associated with related genes, recognised by specific sets of transcription factors.
 - D Within the control element of related genes, the specific numbers of transcription factors binding to the control element will enable related genes to be expressed.
- 15 Gel electrophoresis was performed using DNA samples of gene X isolated from equal number of cells from a human embryo after 2 days, 4 days and 8 days of development.



What kind of gene regulation is illustrated by the results of this gel electrophoresis?

- A DNA demethylation
- B Histone deacetylation
- C Transcriptional activation
- D Gene amplification

- 16 Fruit flies (*Drosophila*), homozygous for long wings, were crossed with flies homozygous for vestigial wings. The F_1 and F_2 generations were raised at three different temperatures.

At each temperature, the F_1 generation all had long wings.

The table shows the results in the F_2 generation.

Temperature / °C	Result
21	$\frac{3}{4}$ long wings, $\frac{1}{4}$ vestigial wings
26	$\frac{3}{4}$ long wings, $\frac{1}{4}$ intermediate wing length
31	all long wings

Which statement explains these results?

- A Heterozygous flies have vestigial wings only at 21°C or below but have long wings at 31°C or above.
 - B Long wing and vestigial wing illustrate codominance at 26°C.
 - C Long wing is dominant at higher temperatures but vestigial wing is dominant at lower temperatures.
 - D Vestigial wing is recessive but causes a vestigial wing phenotype only at lower temperatures.
- 17 In mice, the gene for “dappled” coat (D) and its recessive allele for “plain” coat (d), are located on the X chromosome. The gene for “straight” whiskers (W) and its recessive allele for “bent” whiskers (w), are autosomal.

A male mouse with plain coat and bent whiskers was mated on several occasions to the same female and the large number of offspring consisted of males and females in equal numbers in all possible combinations of phenotypes,

What is the genotype of the female parent?

- A $X^D X^D WW$
- B $X^D X^d WW$
- C $X^D X^D Ww$
- D $X^D X^d Ww$

18 Three gene loci in mice are shown below.

Locus 1 Coat colour	Locus 2 Tail appearance	Locus 3 Coat appearance
agouti albino	kinky straight	non-frizzy frizzy

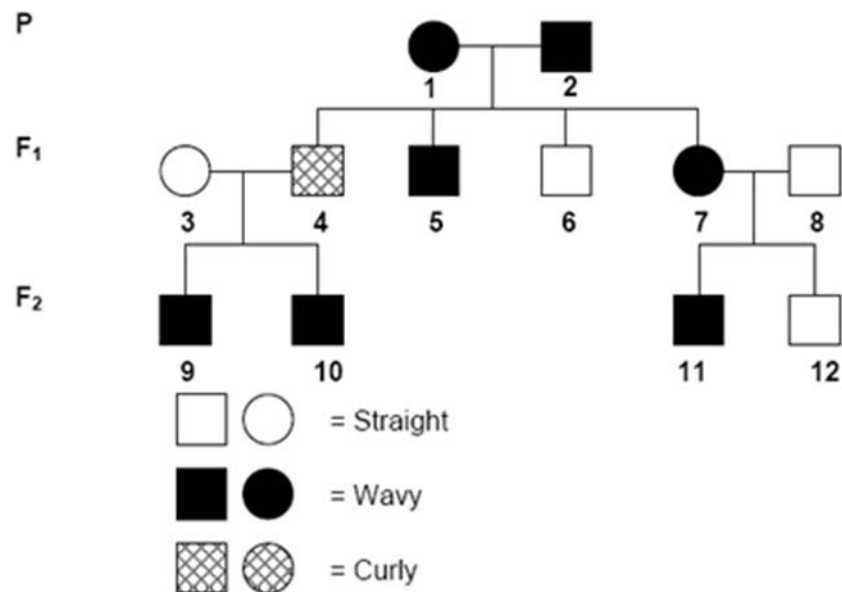
Crosses involving two loci at a time were set up and their outcomes are shown in the table below.

Parents (pure breeding)	F1	Offspring of a test cross of the F1
cross 1 agouti, non-frizzy coat x albino, frizzy coat	agouti, non-frizzy coat	agouti, non-frizzy coat 44 albino, frizzy coat 46 agouti, frizzy coat 5 albino, non-frizzy coat 5
cross 2 agouti, straight tail x albino, kinky tail	agouti, kinky tail	agouti, straight tail 23 albino, kinky tail 27 agouti, kinky tail 24 albino, straight tail 26

Which of the following statement is **TRUE** about **cross 1** and **cross 2**?

	Cross 1	Cross 2
A	The frequency of crossing over between locus 1 and locus 3 is 10%.	The agouti coat and kinky-tailed offspring of the test cross of the F1 are heterozygous at both loci.
B	Locus 1 and locus 3 undergo independent assortment.	The albino coat and straight-tailed offspring of the test cross are pure breeding.
C	Locus 1 and locus 3 are located on the same chromosome.	The F1 mice were test crossed with agouti coat and straight-tailed mice.
D	The interaction between locus 1 and locus 3 is an example of epistasis.	Locus 1 and locus 2 undergo independent assortment.

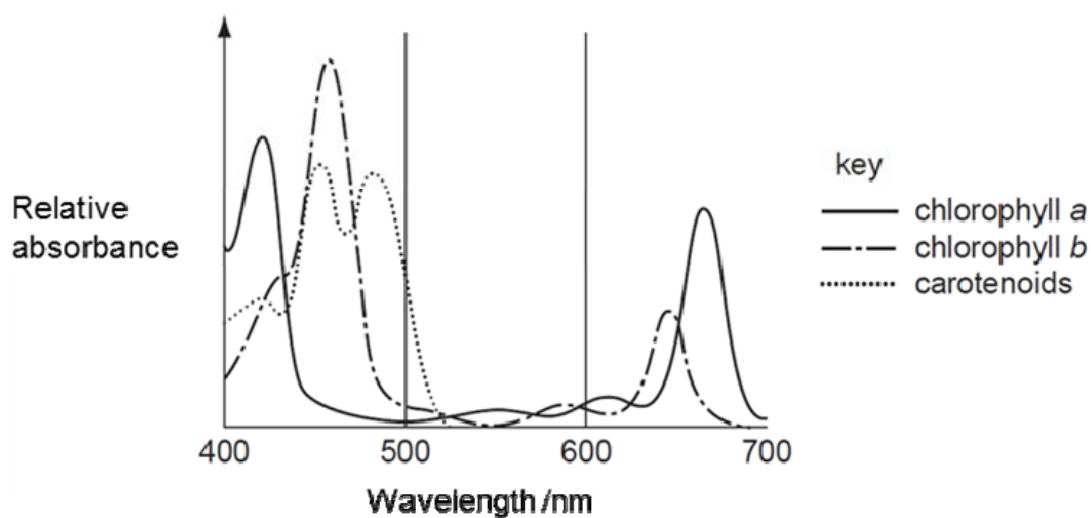
19 The pedigree below shows the inheritance of type of hair.



Which of the following statements are **TRUE**?

- I One of the parents of individual 2 may not always have the same phenotype as individual 2.
 - II If individual 10 married someone with wavy hair, the first child would have wavy hair.
 - III If individual 6 married a woman with straight hair, all of the offspring would have straight hair.
 - IV If individual 7 married a man with curly hair, the first child would have curly hair.
- A** I and III only
B I and IV only
C II and III only
D II and IV only

- 20 The graph shows the absorption spectra of some pigments found in chloroplasts.



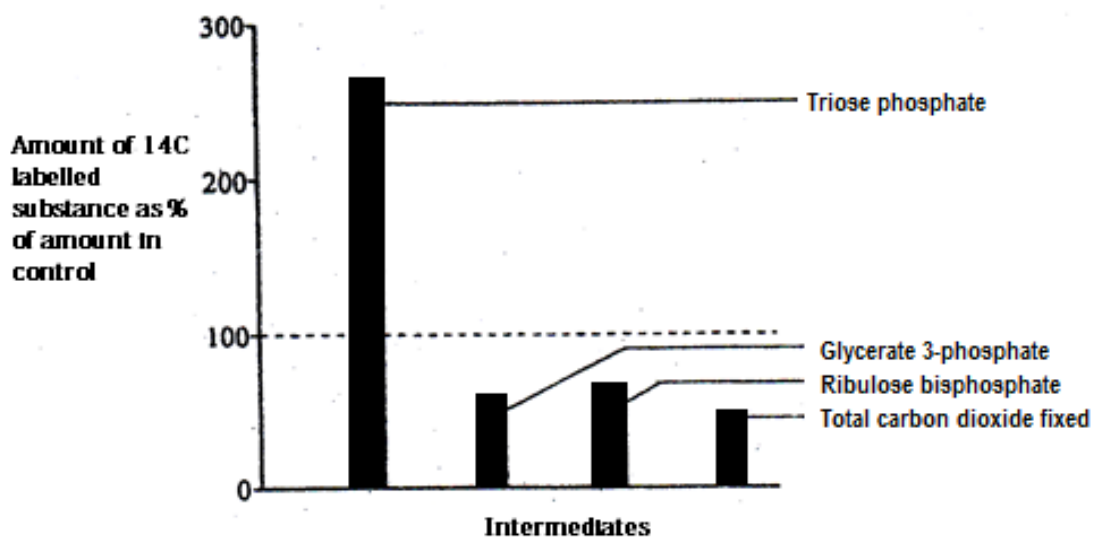
Which statement is **NOT** correct?

- A Having several pigments rather than one increases the efficiency of photosynthesis.
- B Photosynthesis will be fastest when exposed to red light as red light has higher energy than blue light.
- C Prior to leaf fall, chlorophyll is broken down, leaving carotenoids which makes leaves look yellow or red.
- D Most leaves are green as chlorophyll absorbs light in the blue and red regions of the spectrum.
- 21 Removal of the source of carbon dioxide from photosynthesising chloroplasts results in rapid changes in the concentration of certain chemicals. Which one of the following represents the correct combination of concentration changes?

	ATP	ribulose biphosphate	glycerate-3-phosphate
A	increases	increases	decreases
B	increases	decreases	increases
C	decreases	increases	decreases
D	decreases	decreases	increases

- 22 An experiment was conducted to test the properties of a chemical G on the photosynthetic capabilities of a unicellular alga, *Chlorella*. An illuminated suspension of the alga was treated with carbon dioxide labelled with ^{14}C in the presence of an unknown chemical G. The light was switched off and the amount of radioactivity present in some intermediates was determined after 10 minutes in the dark.

A control suspension of alga without chemical G being added was treated in exactly the same manner. The bar chart below shows the amount of radioactivity in these intermediates in the alga with chemical G added as a percentage of the intermediates in the control alga.



Which option **CORRECTLY** describes the action of chemical G?

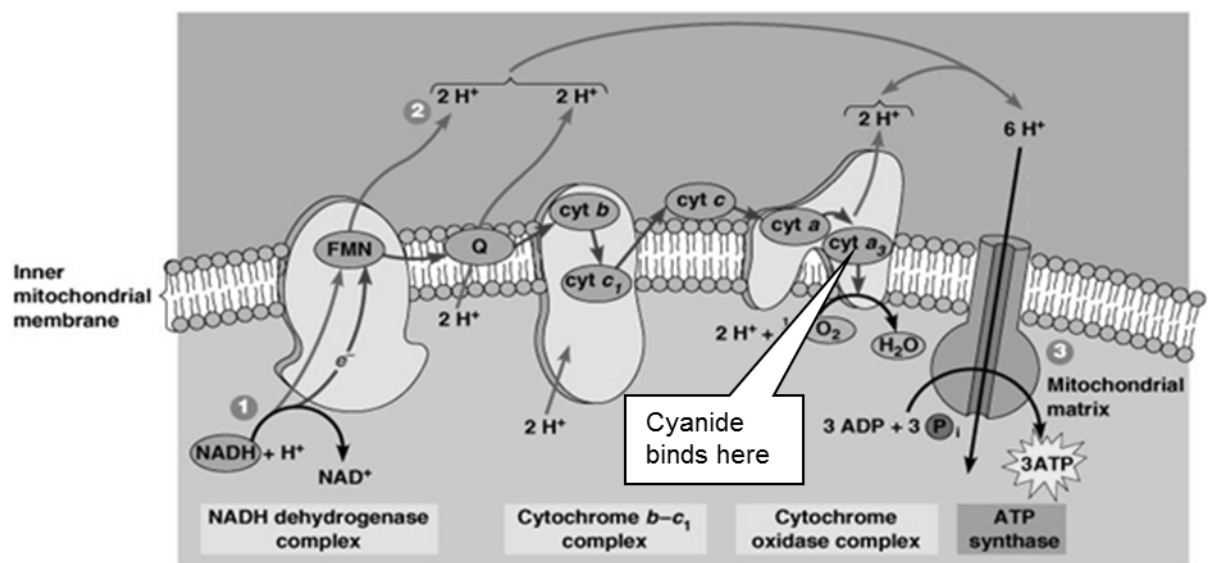
- A G binds to NADPH produced in light reactions and prevent its oxidation process.
- B G competes with triose phosphate for the active site of the enzyme that converts triose phosphate into hexose phosphate.
- C G inhibits the ribulose biphosphate carboxylase enzyme, preventing carbon fixation from taking place efficiently.
- D G prevents the regeneration of ribulose biphosphate at the stage after triose phosphate was formed.

23 Which one of the following substances, when added, would directly result in a decline in ATP production in glycolysis?

- I A chemical that would bind to NAD^+ irreversibly and induces its reduction to NADH.
- II An inhibitor that has a similar structure to glucose but cannot be broken down by respiratory enzymes.
- III A chemical that creates an anaerobic environment by combining with oxygen
- IV A reagent that binds to the active site of ATPase permanently

- A I and II only
- B III and IV only
- C I, II and IV only
- D All of the above

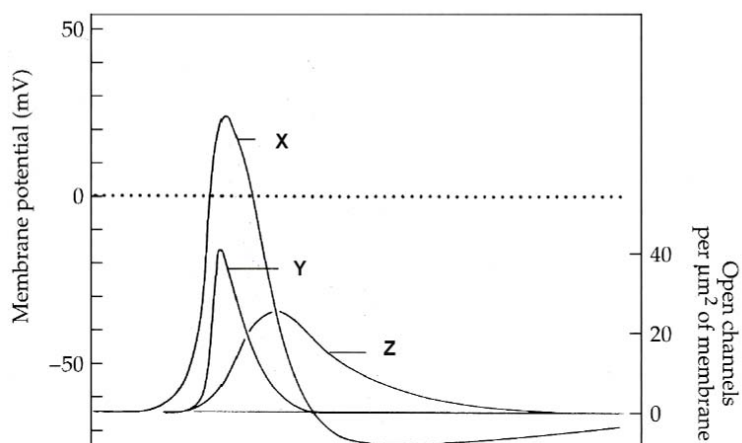
24 Cyanide is an inhibitor that binds irreversibly with the enzyme cytochrome oxidase in the electron transport chain. The diagram below shows the position where cyanide binds.



Which statement is **TRUE** of its effect on cellular respiration?

- A It prevents cells from breaking down glucose.
- B It prevents all synthesis of ATP in the cell.
- C The cell's demand for oxygen would decrease.
- D NAD^+ would still be regenerated at the electron transport chain.

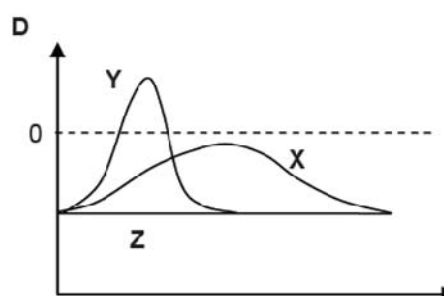
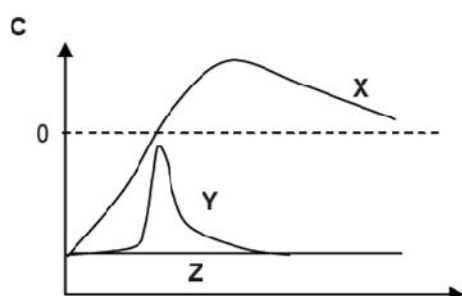
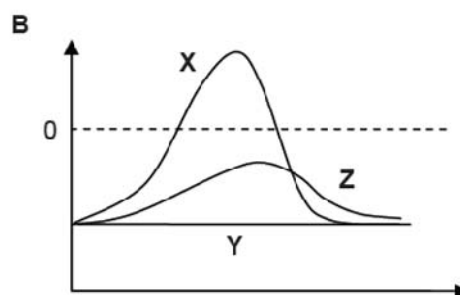
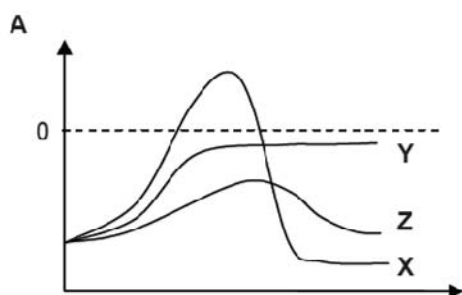
For Questions 25 and 26 refer to the figure below.



- 25 The values of Y and Z affect the value of X in an action potential. What do X, Y and Z in the above figure represent?

	X	Y	Z
A	Change in membrane potential	Sodium ion permeability	Calcium ion permeability
B	Sodium ion permeability	Potassium ion permeability	Calcium ion permeability
C	Change in membrane potential	Sodium ion permeability	Potassium ion permeability
D	Sodium ion permeability	Calcium ion permeability	Potassium ion permeability

- 26 Maurotoxin is a neurotoxin released by scorpions that blocks the pore of the voltage-gated K^+ channel in neurons. Choose from the following graphs to represent how the above diagram will change upon addition of maurotoxin to a neuron.



27 Which of the following statements about diabetes is **FALSE**?

- A In Type 1 diabetes, insulin receptors are absent.
- B In Type 2 diabetes, beta cells of the islets of Langerhans are normal.
- C In Type 1 diabetes, beta cells of the islets of Langerhans are not functional.
- D In Type 2 diabetes, insulin receptors are defective.

28 Which of the following **CORRECTLY** describes the action of a trimeric G-protein?

- A G-protein is phosphorylated and activated by receptor tyrosine kinase.
- B G-protein has GTPase activity, hydrolysing GDP to GTP.
- C Presence of an intracellular signal molecule activates the G-protein coupled receptor, leading to activation of G-protein.
- D Activated subunits of the G-protein travel along the cell surface membrane to activate adenylyl cyclase.

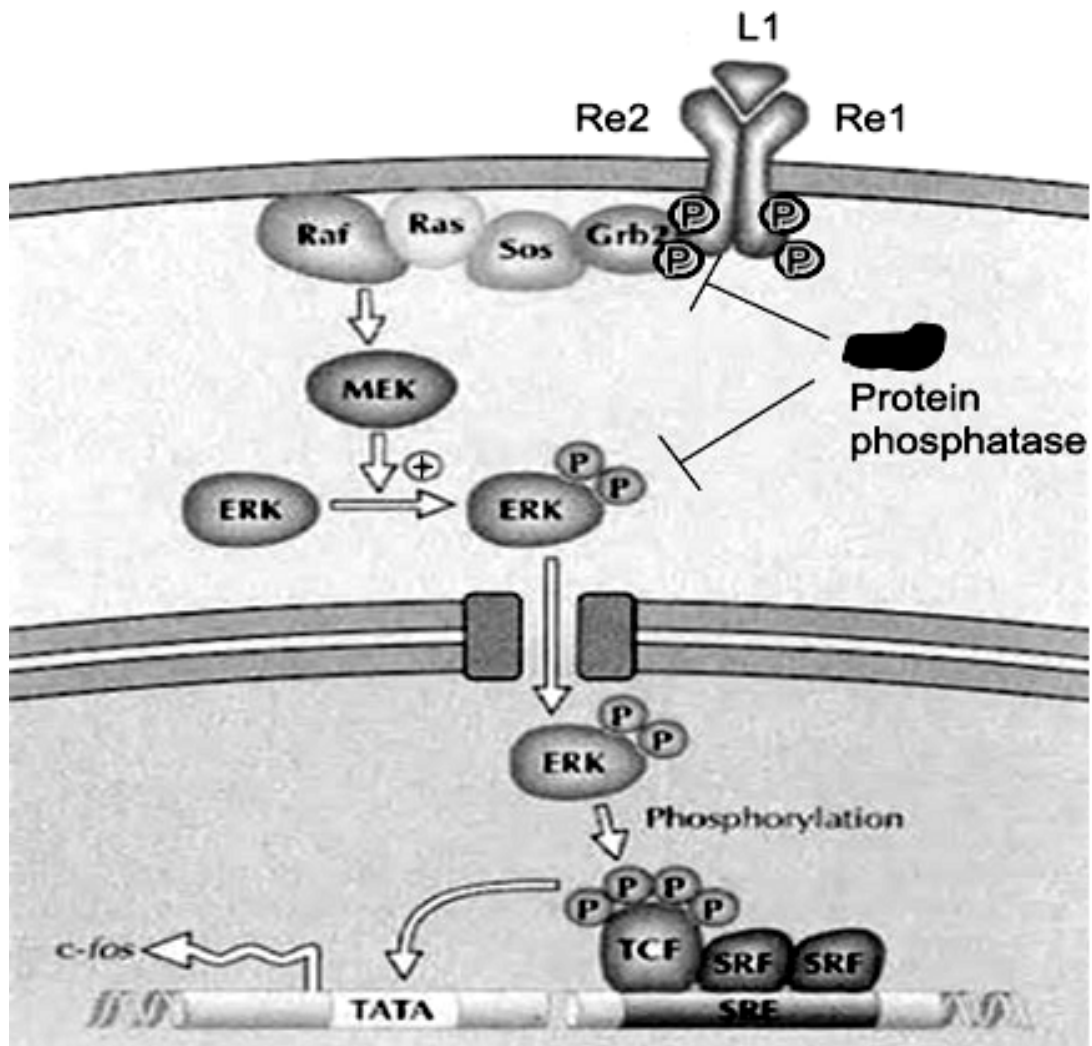
29 A scientist is investigating the effects of Poison T on the cell signalling pathway of glucagon. It is found that Poison T is lipid soluble and diminishes the effect of glucagon. The levels of cAMP were also low in the cell.

Which of the following are possible statements that explain the effects of Poison T?

- I Poison T binds directly to proteins in the cytoplasm.
- II Poison T prevents G protein from hydrolysing GTP.
- III Poison T inactivates the enzyme adenylyl cyclase.
- IV Poison T prevents signal amplification by binding competitively to protein kinase A.

- A I and III only
- B II and IV only
- C I, II and IV
- D I, III and IV

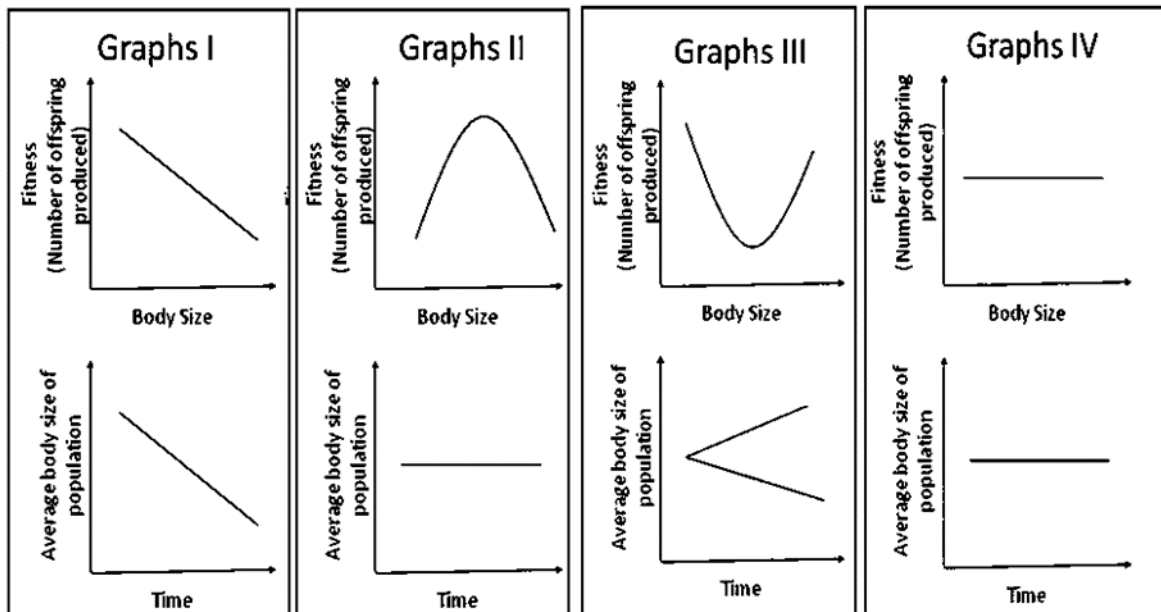
30 The diagram below shows the cell signalling pathway involving a growth factor receptor.



From the given diagram, which step is involved in the role of signal amplification?

- A Binding of L1 to Re1 and Re2.
- B Auto-phosphorylation of Re1 and Re2.
- C Phosphorylation of ERK by MEK.
- D Dephosphorylation by protein phosphatase.

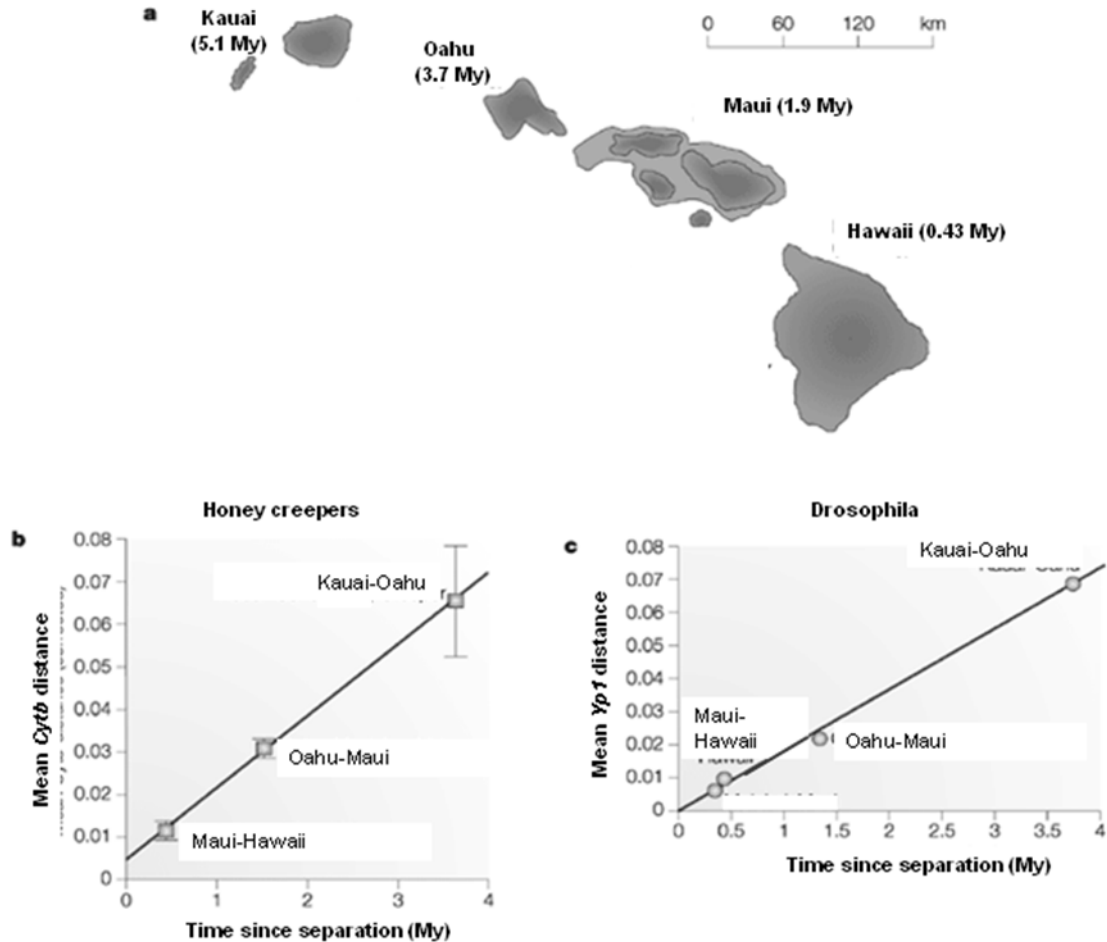
- 31 The different forms of natural selection can be distinguished according to their effect on the body size of the pink salmon (*Onchorhynchus gorbuscha*).



Which of the following describes the **CORRECT** form of natural selection for each of the following sets of graphs?

	Graphs I	Graphs II	Graphs III	Graphs IV
A	Disruptive selection	Directional selection	Stabilizing selection	No selection
B	No selection	Stabilizing selection	Directional selection	Disruptive selection
C	Directional selection	Stabilizing selection	Disruptive selection	No selection
D	Directional selection	Disruptive selection	No selection	Stabilizing selection

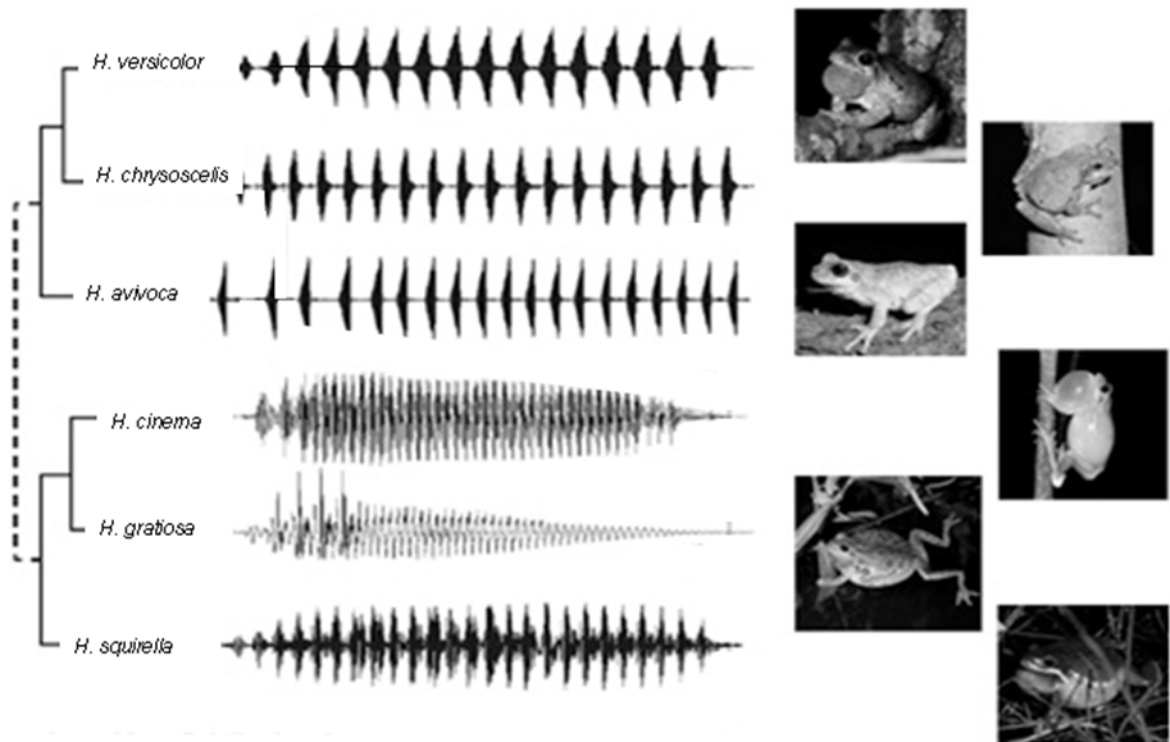
- 32 The volcanic islands that were formed millions of years ago, range from Kauai (the oldest) to Hawaii (the youngest). *Cytb* gene from honey creepers and *Yp1* gene from *Drosophila* were analysed for divergence.



Which of the following statement is **INCORRECT**?

- A Geographical isolation prevented colonization of newly formed islands.
- B There is a positive linear correlation between genetic distance and island age.
- C *Cytb* gene and *Yp1* gene are chosen because they are essential genes.
- D Genetic drift is a factor that contributes to the increase in the mean genetic distance.

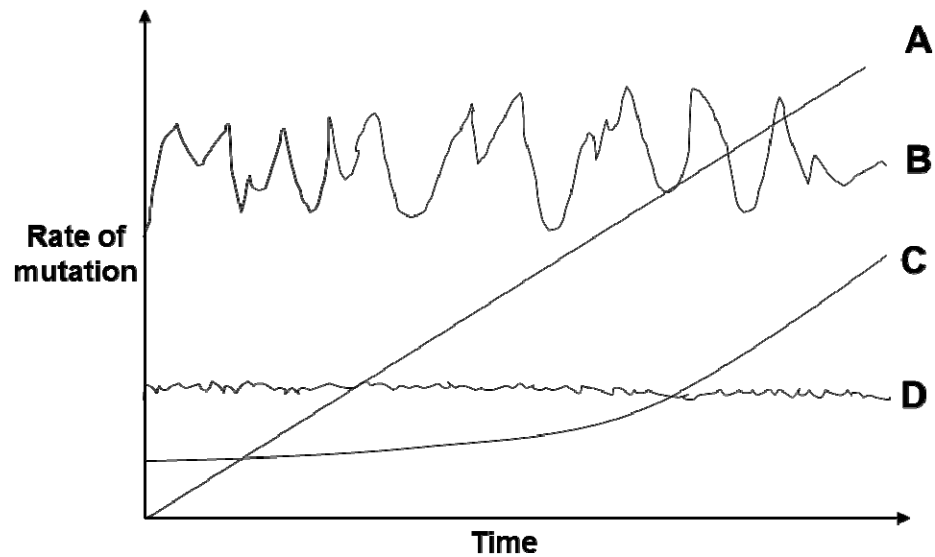
- 33 The calls of six different species of frogs belonging to the *Hyla* genus are recorded and shown.



Which of the following can be inferred from the chart?

- I Frogs with more similar call patterns are more closely related.
 - II A frog species can be identified by looking at the duration, intensity, and frequency of the call.
 - III The call of each species of frog affects their survivability.
 - IV These calls are a form of isolation mechanism.
- A I and II only
- B I, II and IV only
- C I and III only
- D All of the above

- 34 Which pattern of mutation rate would be most helpful if one desires to use a gene as a molecular clock to determine evolutionary relatedness of species that are closely related to each other?



- 35 As part of the procedure to produce recombinant proteins in *E. coli*, you are asked to insert the gene encoding for the MAL protein into the pBN-Z vector. The restriction sites and selectable markers on the vector are shown below.



If the gene for MAL protein were to be inserted into Lac Z site, what should be added to the agar plate in order to screen for recombinant clones and how would the recombinant clones appear?

	Chemicals to be added		Colour of colonies
A	Ampicilin	X-gal	Blue
B	β -galactosidase	X-gal	Blue
C	Ampicilin	X-gal	White
D	β -galactosidase	lactose	White

36 What is the key reason for using a greater range of probes and restriction enzymes in DNA fingerprinting?

- A** It permits more regions of the DNA to be analysed so as to reduce the possibility that two individuals' DNA would produce the same banding pattern.
- B** It is necessary for the creation of unique DNA fingerprints from two individuals' DNA that are significantly different in sequence.
- C** It increases the likelihood that one of the probes will bind to the polymorphic region of the DNA after the latter is cut by the restriction enzymes.
- D** It allows all the DNA bands produced via restriction enzymes cutting to be detected so as to give a more accurate DNA fingerprint.

37 Some of the goals of the Human Genome Project are:

- To determine the sequence of the entire human genome
- To identify all the genes in the human genome
- To find the locus of all the genes on the 46 human chromosomes

Which of the following are ethical concerns arising from the goals stated?

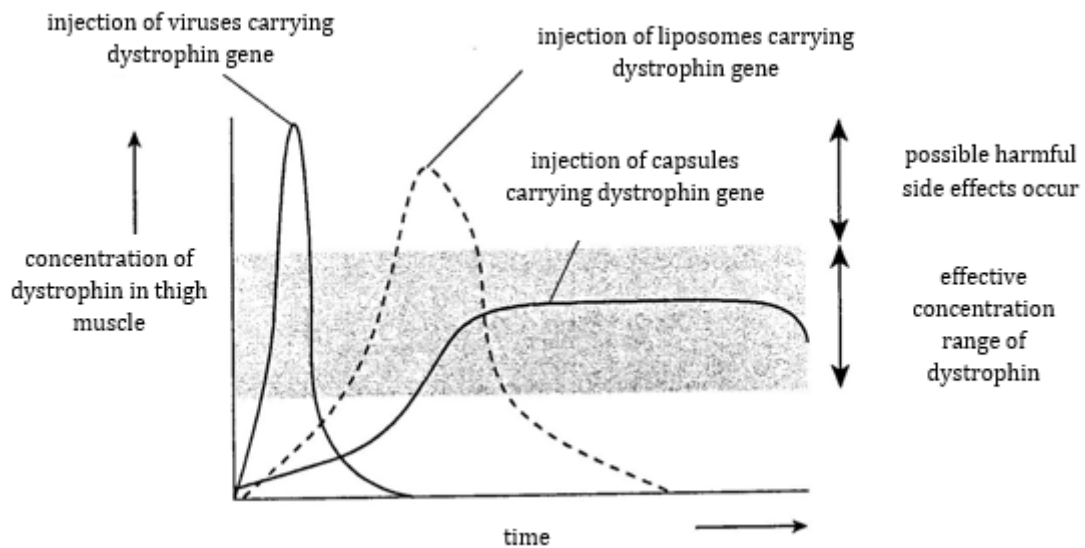
- I** Anthropologist tracing the ancestry of human populations.
- II** Parents choosing embryos for implantation only after tests for acceptable genes.
- III** Insurance company offering cheaper rates to people with genetic disposition to fewer diseases.
- IV** Scientists developing tests for only some disease causing genes.
- V** Genetic counsellors giving advice to people who are genetically pre-disposed to risks.

- A** II and III
- B** III and IV
- C** I and V
- D** IV and V

- 38 Totipotency is demonstrated when _____.
- A cancer cells give rise to heterogeneous cell types.
 - B an isolated plant cell develops into a normal adult plant.
 - C a hematopoietic stem cell differentiates into a lymphocyte.
 - D an embryonic stem cell divides and differentiates.
- 39 Which of the following genetic modifications would **NOT** decrease the quantity of chemicals sprayed onto crop plants by farmers?
- A Fungal resistance
 - B Herbicide resistance
 - C Insect resistance
 - D Viral resistance

- 40 Duchenne muscular dystrophy (DMD) is a lethal X-linked human genetic disease caused by the absence of the protein dystrophin in muscle fibres.

Gene therapy experiments were conducted to compare the effectiveness of three different vectors for introducing a corrective gene coding for dystrophin. Each of the three vectors, virus, liposome and capsule, carrying normal copies of the dystrophin gene was injected into the thigh muscle tissue of DMD patients. The results are shown in the graph.



Which is a viable advantage of the capsule vector over the viral and liposome vectors?

- A It produces more dystrophin.
- B It takes effect in more target cells.
- C It takes a shorter time to take effect.
- D It requires less frequent treatments.

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2016 Y6 Preliminary Exam H2

MCQ Answer Scheme

1	A	21	A
2	B	22	D
3	A	23	A
4	B	24	C
5	C	25	C
6	D	26	C
7	D	27	A
8	C	28	D
9	A	29	D
10	C	30	C
11	D	31	C
12	A	32	A
13	A	33	B
14	C	34	D
15	D	35	C
16	D	36	A
17	D	37	A
18	A	38	B
19	A	39	B
20	B	40	D